QSHIELD-FL

A Privacy-Preserving Quantum Federated Intrusion Detection System with Robust Aggregation and SVD-Enhanced Hybrid Learning

Course	Quantum Computing - Course Project
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Project Status	Finalized with SVD-hybrid rank-16 deployment path
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1. Title and Author Information

Title: QSHIELD-FL: A Privacy-Preserving Quantum Federated Intrusion Detection System with Robust Aggregation and SVD-Enhanced Hybrid Learning

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2. Abstract

QSHIELD-FL is a quantum federated intrusion detection system built for binary attack classification on the NSL-KDD dataset. Rather than centralizing sensitive network data, the system trains a hybrid variational quantum classifier across simulated clients, exchanging only model updates. We evaluate it under four scenarios: clean federated training, SignFlip attacked training, Trimmed Mean defended training, and defended training with additional DP noise.

Over Phase 1, the key addition is an SVD enhanced hybrid learning path, which compresses input features before the quantum circuit. We ran a rank sweep across 16, 24, and 32, and selected rank 16 for deployment because it held up best across both defended and defended plus DP conditions. The system includes a frontend and backend dashboard for browsing result folders, loading checkpoints, and running projection aware inference on the SVD path.

We want to be upfront about scope: the federated setup runs in simulation, not across real networked machines. Differential privacy is implemented as Gaussian update noise, which is practical, but not a full epsilon-delta accountant. QKD is a planned future extension, not something running in the current codebase. The results in the report reflect what was actually built and measured.

3. Keywords

Quantum Federated Learning; Variational Quantum Circuit; Intrusion Detection System; Hybrid Quantum Neural Network; Singular Value Decomposition; Byzantine Robust Aggregation; Differential Privacy; SignFlip Attack; Trimmed Mean

4. Introduction

Intrusion Detection Systems (IDS) are meant to identify malicious network behavior from structured traffic records, but modern enterprise and institutional settings rarely store all such records at a single point. Security data is usually spread across multiple labs, endpoints, departments, or network segments. In such cases, centralized training becomes less attractive because moving all raw traffic data to one server can raise privacy concerns, organizational restrictions, and practical maintenance problems. Federated learning is useful in this context because it enables client-local training and global coordination through model updates instead of raw-data pooling [9].

However, federated learning does not solve every problem by itself. Once a central server starts accepting model updates from clients, the training process becomes vulnerable to malicious behavior. A compromised

or adversarial participant may intentionally transmit a corrupted update so that the global model drifts away from a useful decision boundary. This issue is especially important for IDS because the target application is security-sensitive by definition. A defense mechanism therefore needs to exist not only for malicious network traffic at inference time, but also for malicious behavior during model training itself [3]-[7].

Quantum machine learning has emerged as a promising direction for nonlinear learning under compact circuit settings. In practice, the most realistic approach is not to force the entire pipeline into a purely quantum form, but to use a hybrid architecture in which classical computation performs the parts that are naturally suited to tabular data preparation, while the quantum component serves as the compact trainable nonlinear core [2], [12], [19]. This principle is central to QSHIELD-FL. The final model path does not try to encode the entire raw feature space directly into the circuit. Instead, it uses classical feature preparation and SVD-based compression to form a qubit-compatible latent representation before the variational circuit.

The first version of the report proposed the broad idea of a quantum federated IDS, but the technical logic behind the three professor-specified components was not developed strongly enough. The current version fixes that issue by structuring the report directly around the actual implementation: (i) quantum neural network design using a VQC-based hybrid model, (ii) federated architecture with local client training and server-side aggregation, and (iii) privacy and defense mechanisms through SVD, DP-style update noise, SignFlip attack simulation, and Trimmed Mean robust aggregation. The report also reflects the final deployment state of the project, where the frontend and backend use the SVD-enhanced rank-16 model path as the default demo configuration.

4.1 Research Gap

Looking across the existing literature, the problem is not about any single piece being missing. It is about the fact that no one has put all the pieces together in one working system.

Classical IDS work on NSL-KDD is strong on accuracy and preprocessing, but almost all of it assumes data sits in one place [1], [11], [16]. For benchmarking, it works fine. But it does not tell you anything about what happens when training is distributed across multiple clients who may not trust each other.

FL based IDS papers take a step in the right direction, but most of them stop at showing federated training is feasible [8]–[10]. Very few actually test what happens when a client starts sending corrupted updates, or show how a privacy mechanism and a defense rule interact with each other in the same pipeline. They describe the architecture; they do not break it.

The Byzantine defense literature is more specific. Papers on Trimmed Mean, Median Krum, and committee based validation clearly show naive averaging breaks under poisoning [3]–[7]. But these are almost always tested on generic FL benchmarks, not on an IDS task with a quantum model underneath. A defense working in one setting does not automatically transfer.

Privacy literature adds another wrinkle. Federated learning is often described as preserving privacy by default, but it is not accurate. Gradient leakage attacks show updates can leak information about the raw training data, even when the data never leaves the client [15]. DP and secure aggregation help, but the cost to accuracy is rarely measured carefully in an IDS context [8], [13], [18].

On the quantum side, the circuit design and hybrid model papers are solid [2], [12], [19], but they treat security as out of scope. The QFL survey [14] honestly notes that reproducible quantum federated systems built for deployment barely exist yet.

So what is actually missing is a single pipeline covering the full loop: intrusion detection, federated training, adversarial update simulation, robust aggregation, privacy compression, and evaluation ready for deployment, all in one place. QSHIELD-FL is built to be exactly that.

5. Related Work

5.1 Classical IDS on NSL-KDD and Feature / Data Handling

Classical IDS work on NSL-KDD and similar datasets remains an important reference point because it defines what a tabular intrusion-detection pipeline typically optimizes for: feature processing quality, class balance handling, and benchmark accuracy. Dangol and Eaman [1] show that the effect of resampling is not cosmetic; it can materially change detection outcomes, especially when rare attack classes are involved. Iftikhar et al. [11] similarly reinforce that architecture choice and feature treatment have a large impact on final classification quality. Faiaz [16] goes even further by changing the representation space itself and converting NSL-KDD features into an image-like form for CNN-based modeling. These studies matter for QSHIELD-FL because they set the baseline expectation that data handling is a first-class design decision. At the same time, they do not answer the privacy and robustness questions that arise once learning is distributed across multiple clients.

To keep the final evaluation grounded and not purely quantum-centric, lightweight centralized classical reference models were also implemented in the final project stage. These included Logistic Regression, a compact MLP, and a Decision Tree on the same NSL-KDD binary formulation and preprocessing pipeline. Their role was not to replace the federated scenario analysis, but to provide conventional reference points for how standard classical models behave on the same intrusion-detection task.

5.2 Quantum Machine Learning / Quantum Neural Networks for Learning Tasks

Quantum machine learning papers are most useful here when they explain realistic hybrid design patterns rather than abstract claims about quantum advantage. Ma and Tresp [2] motivate the legitimacy of low-rank compression in quantum-oriented learning settings, even though their domain is not IDS. Verdone et al. [12] demonstrate a practical hybrid pattern in which classical preprocessing narrows the input burden before the variational circuit, which closely matches the final philosophy of the SVD-enhanced path in QSHIELD-FL. Gong et al. [19] focus on QCNN/VQC circuit design and show that encoding choice and ansatz depth strongly affect expressivity and convergence behavior. Taken together, these works support the decision to use a compact hybrid VQC instead of attempting a large, purely quantum model over raw tabular features.

5.3 Federated Learning for IDS

Federated IDS studies show that distributed security learning is a natural fit because the data itself is distributed. Hernández-Ramos et al. [9] organize the field and make clear that privacy-preserving intrusion detection cannot be understood only through centralized baselines. Driouch et al. [10] show that distributed IDS is operationally meaningful beyond the standard NSL-KDD setting. Li et al. [21] and other FL systems papers indicate that client heterogeneity and communication overhead are not side issues; they directly affect convergence and deployment practicality. These works justify the use of a client-server learning

architecture in QSHIELD-FL, but they still leave open the question of how such a system behaves when malicious updates are introduced and how a quantum model fits into that flow.

5.4 Attacks and Defenses in Federated Learning (Byzantine / Backdoor / Robust Aggregation)

The defense layer in QSHIELD-FL is motivated by work that treats federated learning itself as an adversarial surface. Xu et al. [3] show that Byzantine resilience can involve governance ideas such as committee validation and not just a different averaging rule. Liu et al. [4] argue that robust aggregation alone is not always enough under adaptive backdoor behavior. Wang et al. [5] directly support the use of Trimmed Mean as a practical robust aggregation strategy. Colosimo and De Rango [6] provide an alternative perspective through Median-Krum style defense. Liu and Zhang [7] show that robust aggregation and privacy perturbation may need to be considered jointly rather than as isolated modules. These papers collectively motivated the threat-model design in this project: use SignFlip to stress the training process, use Trimmed Mean as the primary defense, and discuss limitations honestly instead of presenting one defense rule as universally sufficient.

5.5 Privacy Mechanisms: Differential Privacy, Secure Aggregation, Compression, Leakage

Privacy-related papers make it clear that federated learning should not be treated as automatically private. Yao et al. [13] discuss secure aggregation from the perspective of privacy protection and integrity. Ding et al. [15] show that gradients, even when compressed, can still leak information. El Ouadrhiri et al. [18] combine compression and differential privacy in a way that supports the idea of privacy interventions at multiple points of the pipeline rather than only one. Liu et al. [8] and Li et al. [21] add the practical angle by connecting privacy and communication efficiency. These studies directly influenced the decision to include DP-style Gaussian update noise and to treat SVD-based compression as part of the privacy-aware design discussion, while also remaining honest that a full formal privacy accountant is not yet implemented.

5.6 Quantum Federated Learning (QFL) Context

Ballester et al. [14] describe quantum federated learning as a developing area with unresolved issues around stability, resources, and realistic deployment. This is important because it supports the project's final design choice: use a modest, interpretable hybrid quantum core instead of overstating what a quantum pipeline can presently do. It also justifies keeping QKD as a future architectural direction rather than pretending that every privacy-related concept in the project brief is already realized in code.

5.7 Comparison Table

Ref	Category	Dataset / Domain	Main Technical Idea	Security / Privacy / FL Relevance	What It Does Not Solve	Why It Matters for QSHIELD-FL
1	Classical IDS	NSL-KDD, UNSW-NB15	Resampling impact on IDS performance and class balance	Security dataset relevance; no FL or adversarial training	Does not address privacy-preserving distributed training or malicious clients	Motivates careful preprocessing and realistic reading of accuracy claims
2	Quantum ML	Knowledge graphs	Quantum learning with tensor/SVD-	Indirect quantum-model	No federated IDS pipeline,	Supports the legitimacy of

			oriented low-rank formulation	relevance; no FL or IDS	no adversarial robustness	rank-controlled dimensional compression
3	FL Security	General FL	Committee-based Byzantine-resilient FL with validation logic	Strong federated defense relevance	Not a quantum IDS implementation	Supports the adversarial-client narrative and defense-as-pipeline framing
4	FL Security	General FL	Backdoor defense beyond naive robust aggregation	Strong poisoning-defense relevance	Not IDS-specific and not quantum-specific	Justifies careful non-overclaim language around single-rule defenses
5	Robust Aggregation	General FL	Trimmed Mean aggregation rules for Byzantine resilience	Direct defense-rule relevance	Does not provide full-stack IDS deployment	Direct conceptual support for the implemented Trimmed Mean defense
6	Robust Aggregation	General FL	Median-Krum style Byzantine-robust aggregation	Defense benchmarking relevance	Not an IDS-specific end-to-end pipeline	Useful as an alternative defense reference and limitation discussion point
7	Robust + DP	General FL	Robust aggregation compatible with differential privacy	Joint privacy-defense relevance	Not quantum or IDS specific	Supports the defended-plus-DP experimental scenario
8	Privacy / Efficiency	General FL	Quantization/compression and privacy-preserving FL flow	Privacy and communication relevance	No quantum IDS system integration	Supports privacy-utility and communication tradeoff discussion
9	FL-IDS Survey	IDS	Taxonomy and	Direct FL-IDS	Survey rather	Supports

			challenges of FL-based IDS	relevance	than implemented pipeline	motivation for client-local IDS training
10	Distributed IDS	Satellite CAN traffic	Distributed IDS outside standard NSL-KDD setting	Distributed security-system relevance	No quantum or robust FL focus	Shows distributed IDS is meaningful beyond a single benchmark dataset
11	Classical IDS	NSL-KDD	Hybrid SOM-based IDS design	IDS baseline relevance	No FL, no privacy mechanism, no adversarial FL	Provides centralized baseline context for intrusion detection
12	Hybrid QNN	Healthcare classification	Classical compression plus VQC hybrid model	Strong hybrid-QNN design relevance	No FL security or IDS domain	Supports the classical-front-end plus quantum-core design pattern
13	Secure Aggregation	FL networks	Verifiable secure aggregation for privacy/integrity	Privacy-preserving FL relevance	Not an implemented quantum IDS pipeline	Supports future security-channel strengthening discussion
14	QFL Survey	Quantum FL	Foundations, challenges, and future directions of QFL	Strong conceptual quantum-federated relevance	No IDS-specific deployment path	Supports the hybrid and cautiously scoped quantum design choice
15	Privacy Attack	Federated learning	Gradient leakage from compressed updates	Direct privacy-risk relevance	No IDS system integration	Supports the need to avoid overstating privacy from FL alone
16	Classical IDS	NSL-KDD	CNN color-mapping approach for IDS	Accuracy/reference relevance	No FL, no robust training story	Supports the statement that centralized

						accuracy is not enough
17	Communicati on-Efficient FL	General FL	Hierarchical compression for efficient FL	Federated systems relevance	Not a quantum IDS pipeline	Supports discussion of scalability and heterogeneo us-client concerns
18	Privacy + DP	Federated learning	Compression plus differential privacy	Direct privacy-mechanism relevance	Not quantum IDS specific	Supports the SVD-plus-DP discussion as a practical privacy path
19	Quantum Circuit Design	ML classification	QCNN/VQC design and encoding-depth tradeoffs	Strong circuit-design relevance	Not IDS or FL oriented	Supports deeper gate-level circuit explanation in the report
20	Quantum IDS	IDS	Optimized QNN with cryptographic security framing	IDS plus quantum motivation	Weak federated rigor and limited robust FL evaluation	Useful for IDS-specific quantum motivation and overclaim discipline
21	Communicati on-Efficient FL	FL heterogeneo us clients	Efficiency and heterogeneity-aware coordination	Federated deployment relevance	No quantum IDS integration	Supports practical deployment constraints discussion

Table 5.7 keeps one paper per row for traceability. This format is important because the final system does not emerge from a single source. Different papers influence different parts of the pipeline such as preprocessing, hybrid model design, robust aggregation, privacy discussion, and deployment caution.

6. Problem Formulation

Let the dataset be D = {(x_i, y_i)} where x_i ∈ R^d represents a tabular network feature vector and y_i ∈ {0,1} indicates whether the record is Normal or Attack. After one-hot encoding and scaling, the raw feature dimensionality used in the implementation is 122. In the SVD-enhanced path, the raw feature vector is compressed to rank r before entering the hybrid model.

The project objective is to learn a classifier f(x) that predicts y while satisfying three system-level requirements. First, learning must support client-local federated training without direct exchange of raw feature records. Second, the training process must remain resilient when one or more clients send malicious updates. Third, the privacy layer should include practical compression and update perturbation mechanisms while staying feasible within a course-project implementation.

During each federated round, client k receives the current global parameter set W_{global}, performs local optimization on its client partition D_k, and computes a parameter delta $\Delta W_k = W_k - W_{global}$. If the round is attacked, the selected malicious client update is transformed using SignFlip, i.e., $\Delta W_m \leftarrow -\alpha \Delta W_m$, where α is the attack scale. If DP-style perturbation is enabled and the round index has crossed the configured start round, Gaussian noise is added to each update tensor. The server then performs either FedAvg or coordinate-wise Trimmed Mean before updating the global state.

For the SVD-enabled model path, the projection is defined as $Z = (X - \mu)P_r$, where μ is the empirical mean of the training features and P_r is the retained right-singular-vector projection matrix for rank r. This formulation allows the report to connect the privacy-mechanism requirement to an implemented dimensionality-reduction path rather than leaving compression as a purely conceptual extension.

- Client update: $\Delta W_k = W_k - W_{global}$
- Attack model: $\Delta W_m \leftarrow -\alpha \Delta W_m$
- FedAvg: $\Delta W = (1/K) \Sigma_k \Delta W_k$
- Global update: $W_{global} \leftarrow W_{global} + \Delta W$
- DP-style noise: $\Delta W_k \leftarrow \Delta W_k + N(0, \sigma^2 I)$
- SVD projection: $Z = (X - \mu)P_r$

7. Proposed Method / Algorithm

7.1 System Architecture & Final Design

The finalized system can be understood as four connected layers. The first layer is the data layer, where NSL-KDD is loaded, categorical attributes are one-hot encoded, numerical features are scaled, and the resulting arrays are partitioned into client-local loaders. The second layer is the model layer, which contains the baseline HybridQNN path and the finalized HybridSVDQNN path. The third layer is the security layer, where SignFlip, DP-style update noise, and Trimmed Mean aggregation operate on client deltas. The fourth layer is the application layer, where the backend and dashboard expose saved result folders, metadata, checkpoints, and projection-aware inference for the SVD path.

From an implementation perspective, the baseline path remains preserved, while the final deployment path defaults to the SVD-hybrid rank-16 artifacts. This separation is useful because it keeps the original system available for comparison while allowing the improved path to become the active demo configuration.

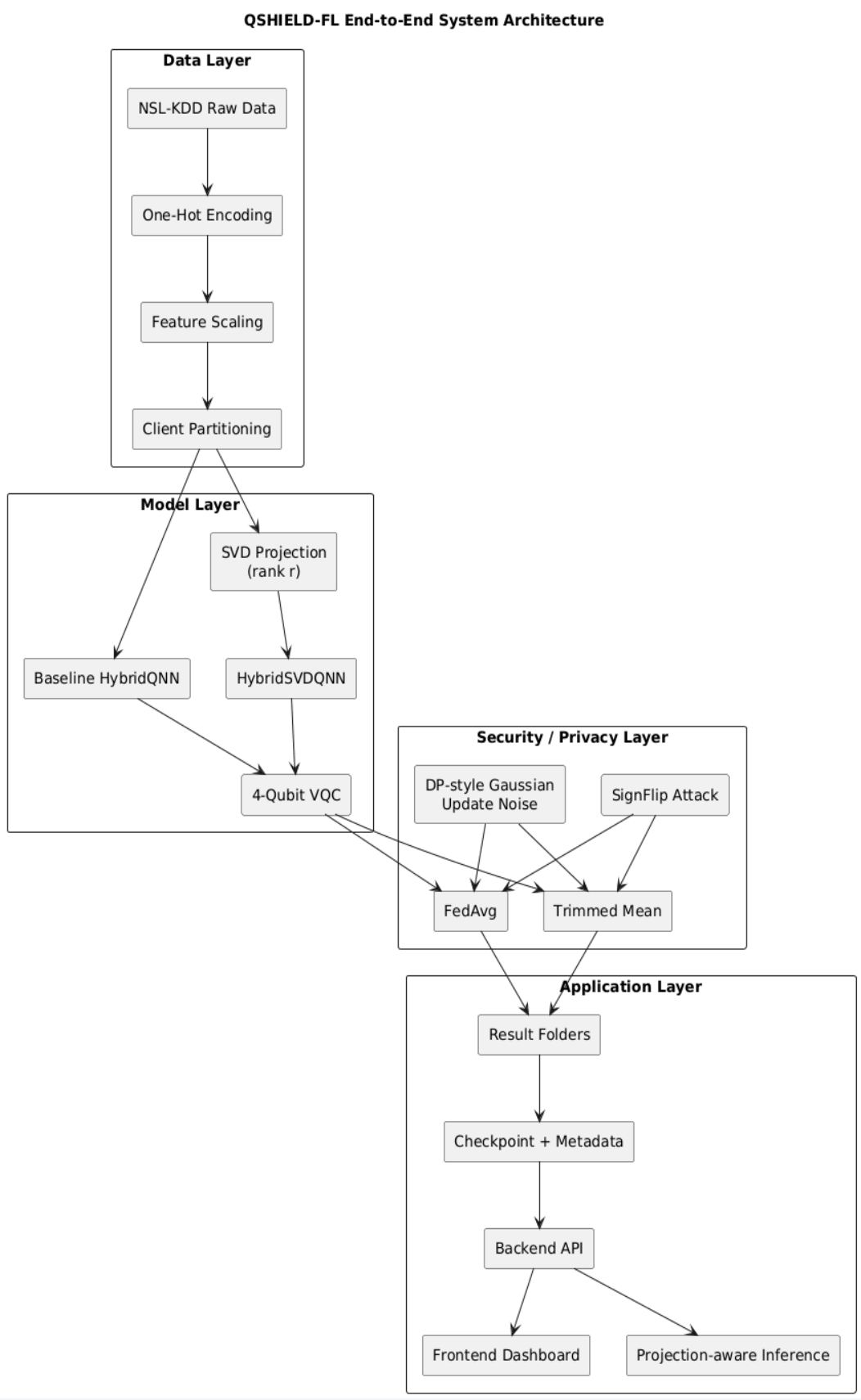


Figure 7.1. End-to-end architecture of QSHIELD-FL showing data preprocessing, client-local training, server aggregation, privacy/defense modules, and dashboard integration.

7.2 Quantum Modelling

The quantum core used in both the baseline and SVD-enhanced variants is a four-qubit variational quantum circuit implemented with PennyLane on the lightning.qubit simulator backend. The circuit does not consume the raw 122-dimensional feature vector directly. Instead, a classical feature extractor maps the current input to a four-dimensional latent vector compatible with the number of qubits. In the SVD path, this input is first compressed to rank r before the classical feature network. This hybrid design choice is technically important because it acknowledges the practical limitation that a small, simulator-friendly qubit budget is not a natural match for raw high-dimensional tabular IDS data.

The embedding block uses AngleEmbedding with Y-axis rotations. Therefore, the four latent values x0, x1, x2, and x3 are injected as RY(x0), RY(x1), RY(x2), and RY(x3) on qubits q0, q1, q2, and q3 respectively. After embedding, the ansatz repeats a three-layer trainable block. In each layer, every qubit receives a three-parameter rotation gate Rot(ϕ, θ, ω), which can be represented in IBM Composer style as an RZ - RY - RZ decomposition. The entanglement pattern is a directional ring: CNOT(0→1), CNOT(1→2), CNOT(2→3), and CNOT(3→0). The readout is the vector of Pauli-Z expectation values on all four qubits, which is then processed by a classical linear output head to produce the final binary logit.

This exact description should be used for the IBM Composer-style figure because it matches the implemented topology. The figure must therefore show four qubits, one Y-angle embedding per qubit, three repeated ansatz layers, each layer decomposing Rot into RZ - RY - RZ, and the wrap-around ring entanglement including the final 3→0 connection. The caption should state clearly that the diagram is a conceptual IBM Composer-style reconstruction of the implemented VQC topology rather than an auto-exported hardware execution circuit.

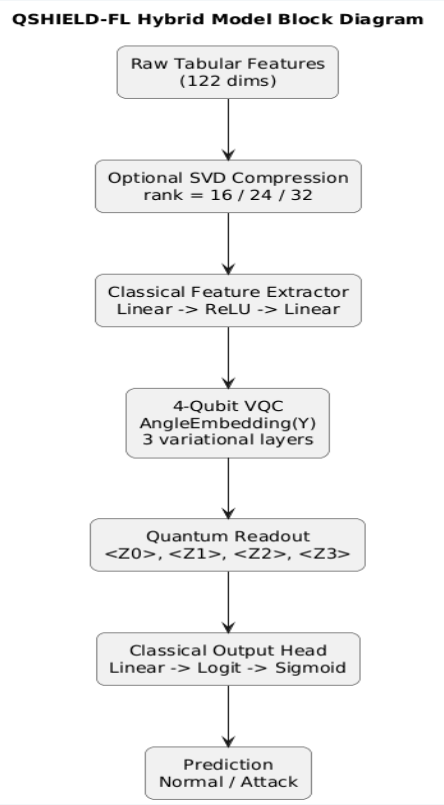


Figure 7.2. Hybrid model block diagram: optional SVD compression, classical feature extractor, 4-qubit VQC, and classical output head.

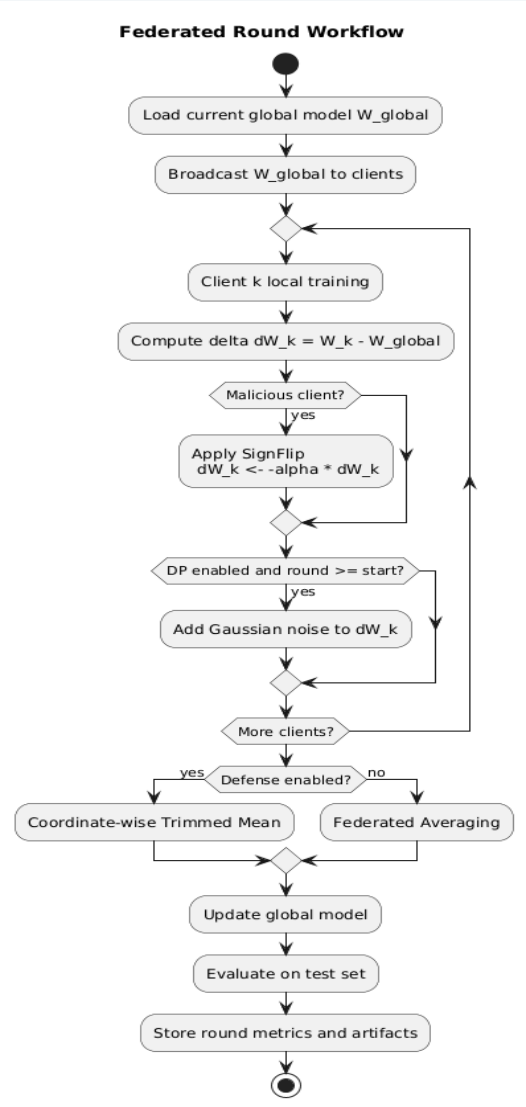


Figure 7.3. IBM Composer-style conceptual reconstruction of the implemented four-qubit VQC with Y-angle embedding, three variational layers, ring CNOT entanglement, and Pauli-Z readout.

7.3 Hybrid Classical-Quantum Extension with SVD

The earlier report stated that a stronger hybrid classical-quantum extension would be developed later, and this final version realizes that extension through SVD-enhanced feature compression. The key idea is not simply to add another block to the network, but to use classical computation where it is more natural and computationally efficient, while preserving the quantum circuit as the nonlinear core. In practical terms, the raw feature vector is centered and projected to rank r before entering the classical feature extractor. This reduces representation size, lowers the burden on the hybrid model, and gives a concrete implementation for the privacy-related compression requirement.

Three SVD ranks were evaluated in the final experimental comparison: 16, 24, and 32. The same VQC topology was kept constant across the sweep so that the effect of dimensional compression could be interpreted cleanly. This is important from a methodology standpoint because it avoids conflating changes in circuit design with changes in feature compression. The final application path uses rank 16 because it provided the best balanced behavior across clean, defended, and defended-plus-DP scenarios.

This figure is a script-generated reconstruction of the implemented QNode topology from the project code, using dummy parameter values only for visualization. It reflects the gate structure used by the default model configuration, but it is not an IBM hardware export or a training-trace snapshot. The implemented readout is based on $\langle Z \rangle$expectation values on all four qubits, which are then passed to the classical output head.

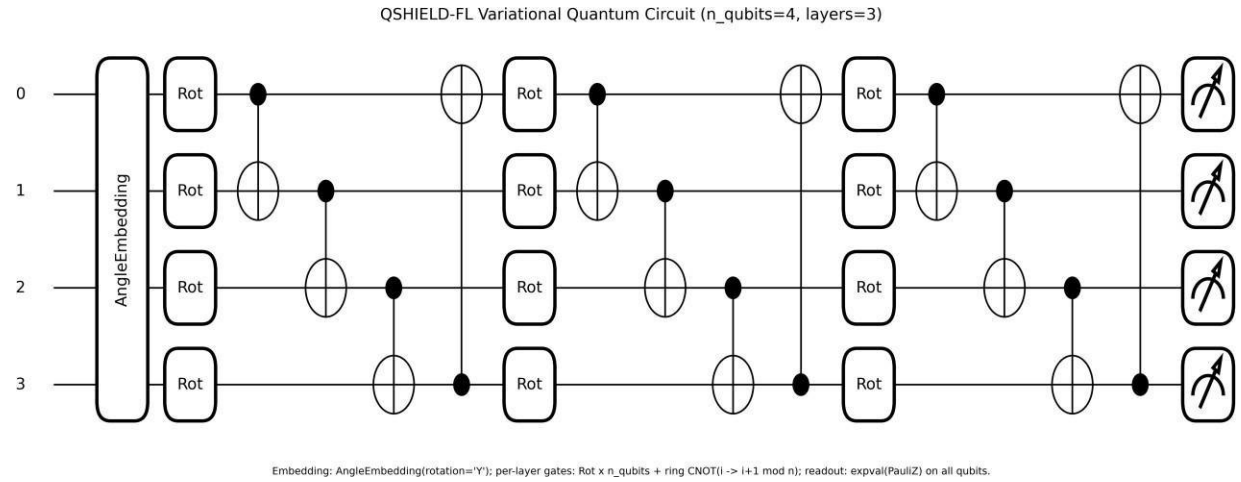


Figure 7.3. QSHIELD-FL implemented variational quantum circuit (default $n_qubits = 4$, $vqc_layers = 3$): AngleEmbedding with Y-rotation, per-layer Rot gates on all qubits, directed ring CNOT entanglement $(i \rightarrow i + 1 \, mod n)$, and Pauli-Z expectation-value readout on all qubits.

7.4 Federated Learning and Security Pipeline

Each federated round follows the same high-level flow. The current global model is copied to each participating client. Local training is then performed on each client partition. After local optimization, the difference between local and global parameters is extracted as the client update. If the scenario is attacked, the selected malicious client has its update transformed by SignFlip. If DP-style perturbation is enabled and the round index is large enough, Gaussian noise is added to the client update. The resulting update set is then aggregated either by standard FedAvg or by coordinate-wise Trimmed Mean. The global model is updated, evaluated on the test set, and the round-level metrics are recorded.

From a security perspective, this organization is more informative than reporting a single final score. It allows the report to show how the same model behaves under four distinct operating conditions: normal training, poisoned training, defended training, and defended-plus-noise training. This scenario-based organization is one of the practical strengths of the final system.

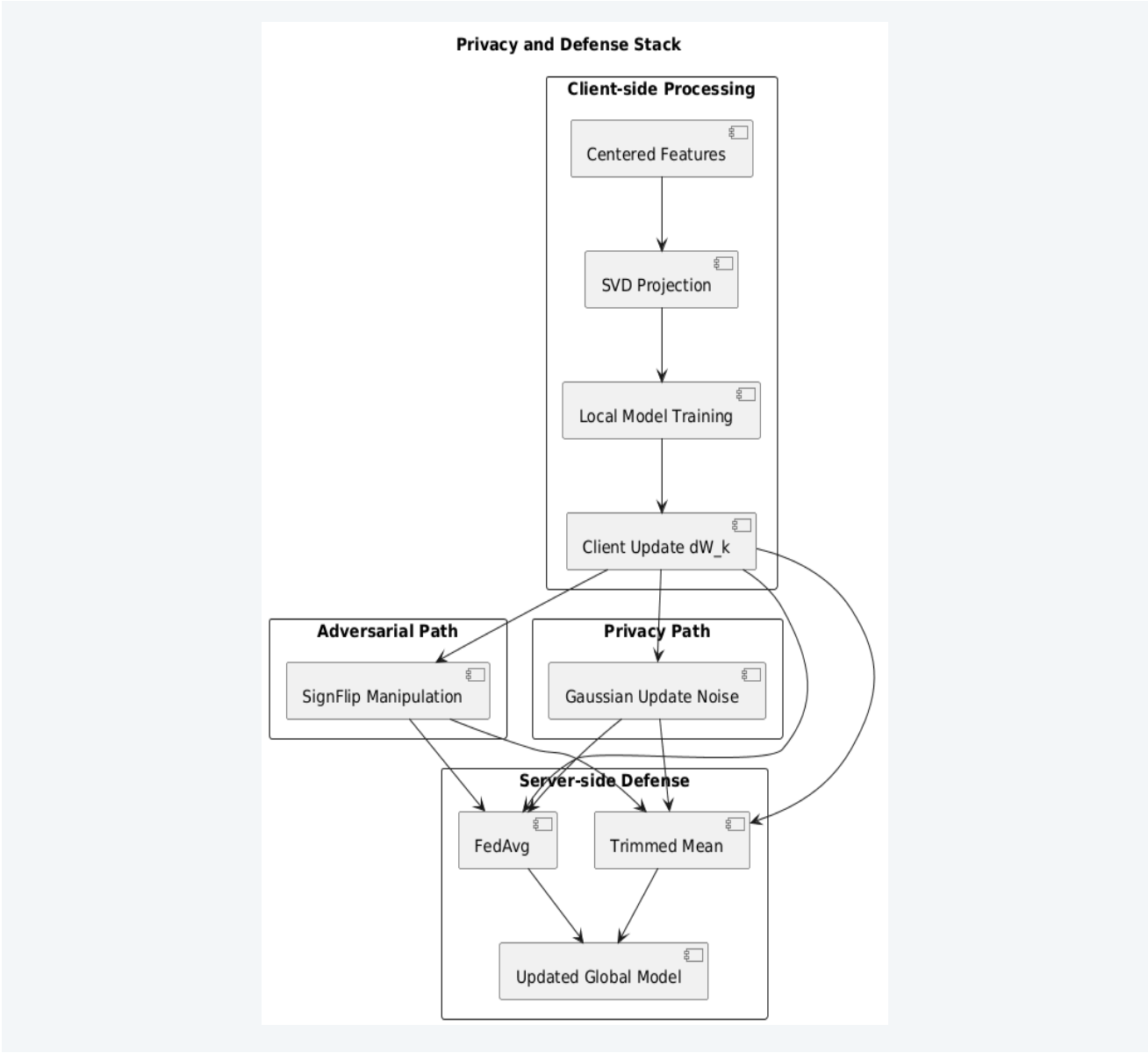


Figure 7.4. Privacy and defense stack showing SVD-based compression, SignFlip attack path, DP-style update perturbation, and Trimmed Mean aggregation.

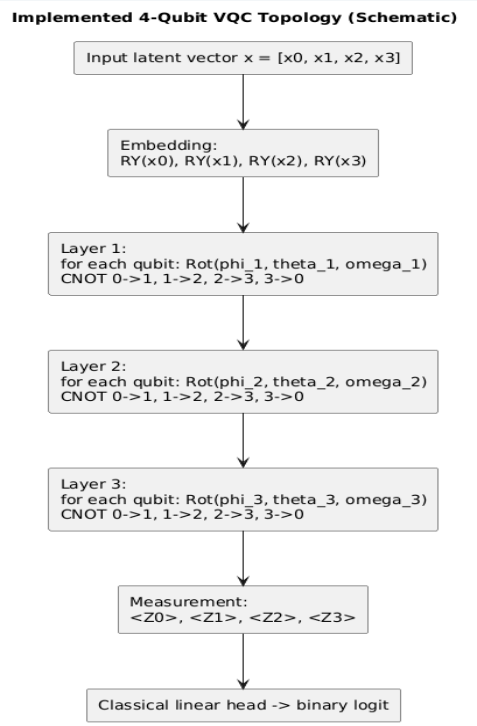


Figure 7.5. Federated round workflow showing local training, delta extraction, optional attack/noise injection, robust aggregation, and global evaluation.

7.5 Novelty and Contributions

The main novelty of QSHIELD-FL is integration-level novelty rather than the invention of an entirely new base learning rule. Trimmed Mean, FedAvg, variational circuits, and Gaussian-noise perturbation are all known building blocks. What is new in this project is the way these components are combined into one evidence-backed, end-to-end student implementation for intrusion detection. The final system joins a hybrid VQC-based IDS, a federated client-server workflow, adversarial update simulation, robust aggregation, privacy-oriented SVD compression, DP-style noise, and a deployment-facing dashboard into a single reproducible pipeline.

The project also adds experimental novelty at the organization level. Instead of reporting only one benchmark score, it evaluates the model under clean, attacked, defended, and defended-plus-DP settings. In addition, the final hybrid path is not chosen arbitrarily; it is selected through an explicit rank sweep over 16, 24, and 32, with rank 16 chosen on the basis of balanced defended performance and clean accuracy.

A third contribution is deployment-oriented reproducibility. The finalized frontend and backend are not generic placeholders; they are explicitly wired to the finalized SVD-hybrid rank-16 folder and checkpoint, while retaining baseline fallback support. This gives the project a concrete demo path rather than leaving the results as detached offline artifacts.

At the same time, the report must stay disciplined about what is not novel or not fully implemented. FedAvg is not novel. Trimmed Mean is not novel. The variational circuit building blocks themselves are standard. Differential privacy is not implemented here as a formal accountant-backed guarantee stack. QKD is not a runtime component in the codebase. Therefore the correct claim is that the project contributes a cohesive, technically grounded integration and evaluation framework rather than a fundamentally new standalone algorithm.

8. Experimental Setup

The experiments were carried out on the NSL-KDD dataset in a binary Normal-versus-Attack setting. The system uses CPU execution with PennyLane lightning.qubit as the simulator backend and PyTorch for the classical optimization path. The final SVD-rank comparison was executed using the cached 20% subset mode in order to keep runtime manageable while still preserving a meaningful comparative signal across scenarios.

Four scenarios were defined and evaluated for each selected model path. Clean training uses no malicious client and no robust defense. Attack training uses SignFlip without defense. Defended training uses SignFlip together with Trimmed Mean. Defended-plus-DP adds DP-style Gaussian noise to the defended pipeline after the configured start round. Clean and attack scenarios use three clients, whereas the defended and defended-plus-DP scenarios use nine clients to stress the aggregation rule under a broader client set.

Parameter	Final Setting
Dataset	NSL-KDD
Target	Binary classification (Normal vs Attack)
Device	CPU
Quantum backend	PennyLane lightning.qubit
Batch size	256
Learning rate	0.003
Rounds in SVD rank sweep	10
Local epochs	2
Qubits	4
Variational layers	3
Attack model	SignFlip (scale = 5.0)
Defense	Trimmed Mean (trim_ratio = 0.25)
DP-style noise	Gaussian update noise (sigma = 0.003, start round = 8)
SVD ranks tested	16, 24, 32

The final baseline comparison uses the older baseline folder as reference. A fairness note is necessary here: the baseline folder uses 20 rounds, whereas the SVD rank sweep uses 10 rounds. Therefore the report treats baseline-versus-SVD superiority claims cautiously and focuses more strongly on relative behavior among the SVD variants and on scenario-level stability.

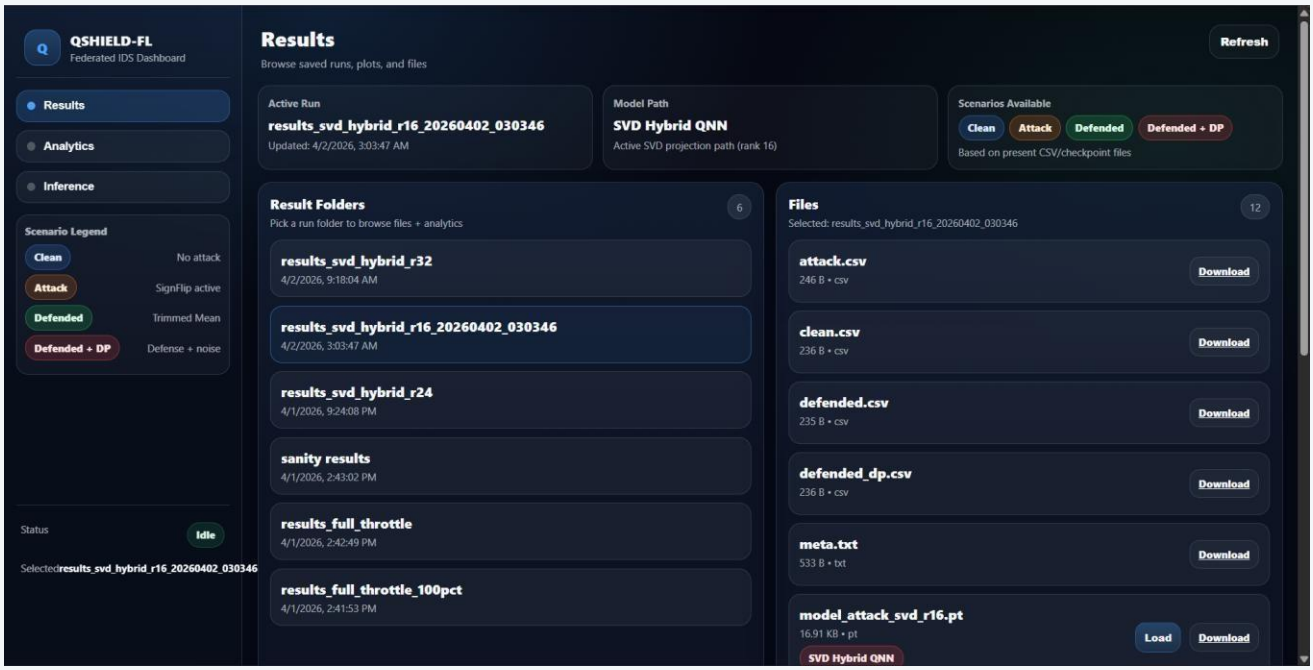


Figure 8.1. Admin/dashboard screenshot showing the finalized SVD-r16 default folder and loaded checkpoint status.

8.1 Classical Reference Baseline Setup

In addition to the federated hybrid experiments, three lightweight centralized classical reference models were also evaluated in order to place the final hybrid results in context. These models were Logistic Regression, a compact MLP, and a Decision Tree. Unlike the hybrid QSHIELD-FL pipeline, these baseline runs were not executed through the full clean/attack/defended/defended-plus-DP federated scenario loop. Instead, they were trained as conventional centralized reference models on the same NSL-KDD binary task using the same preprocessing pipeline.

The purpose of these baselines was not to claim a protocol-matched comparison with the federated hybrid system, but to provide a standard non-quantum reference for pure classification performance. For this reason, these models were run on the full dataset rather than the 20% cached subset used for the SVD rank sweep. This gives them a fair chance as conventional classical baselines while keeping the main system comparison centered on the federated hybrid pipeline.

Table 8.2. Centralized classical reference baseline performance on NSL-KDD

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.7511	0.6476	0.9259	0.7621
MLP	0.7794	0.6788	0.9261	0.7834
Decision Tree	0.7833	0.6849	0.9204	0.7854

9. Results and Comparative Analysis

Model Path	Clean	Attack	Defended	Defended + DP
Baseline HQNN (20 rounds)	0.8053	0.4308	0.7857	0.7614
SVD rank 16 (10 rounds)	0.8072	0.4308	0.7721	0.7741
SVD rank 24 (10 rounds)	0.7905	0.4308	0.7719	0.7416
SVD rank 32 (10 rounds)	0.7935	0.5661	0.7517	0.7370

The updated project results are meaningful in two different ways. First, they show that the new SVD-hybrid path is not merely a cosmetic addition; it changes the operating profile of the model and therefore has to be evaluated rather than assumed. Second, they provide a rational basis for selecting a final deployment path instead of leaving the model choice open-ended.

Among the SVD variants, rank 16 gives the strongest clean accuracy at 0.8072 and also the strongest defended-plus-DP result at 0.7741. Rank 24 remains close in defended performance but falls behind in both clean and defended-plus-DP settings. Rank 32 behaves differently: it provides the best attack-only result at 0.5661, which suggests that larger retained dimensionality changes the poisoned-training dynamics under naive aggregation, but this does not translate into the best defended or defended-plus-DP behavior. Since the project is explicitly about built-in adversarial defenses, the defended scenarios matter more for the final model choice than the attack-only case.

For this reason, rank 16 is selected as the final model. It gives the best overall balance across clean and defended conditions and is therefore the most suitable rank for the final application path. The dashboard integration was updated accordingly so that the finalized rank-16 folder and the clean rank-16 checkpoint are loaded by default.

Comparison with the older baseline should be interpreted carefully because the baseline and SVD rank sweep use different round counts. Even so, the final-rank comparison shows that rank 16 is at least competitive with the baseline in clean performance and slightly stronger in the defended-plus-DP scenario, which supports its use as the finalized hybrid path while still leaving room for future, more controlled repeated-seed comparisons.



Figure 9.1. Accuracy-vs-round curves for clean, attack, defended, and defended-plus-DP under the finalized rank-16 SVD-hybrid path.

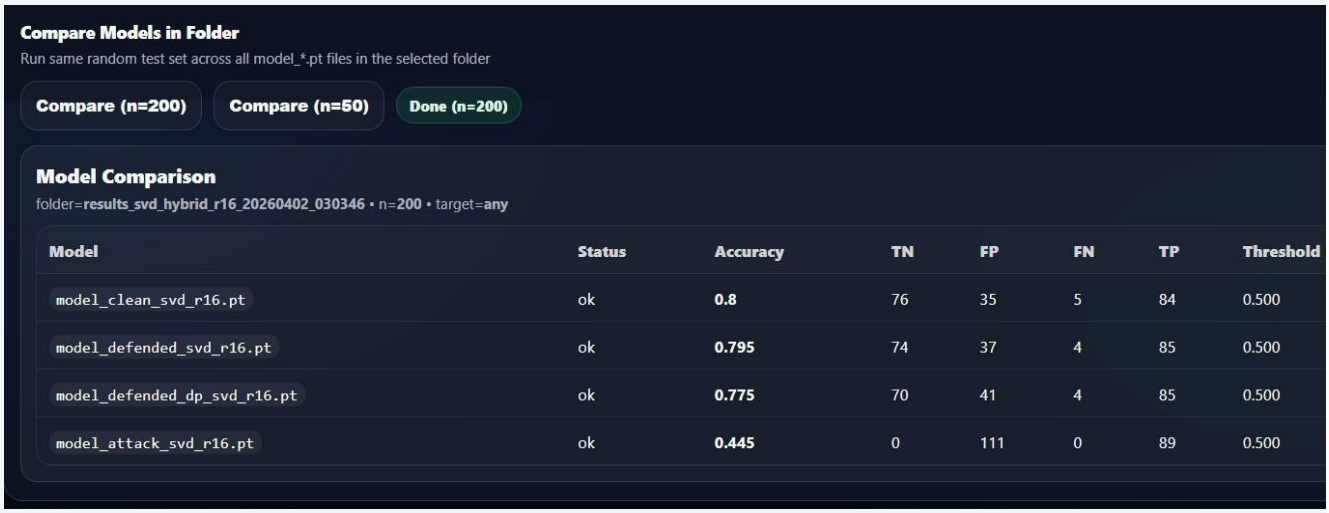


Figure 9.2 Dashboard-based comparison of all SVD-r16 checkpoints in the finalized result folder using a common test batch. The interface reports accuracy, confusion-matrix counts, and threshold for the clean, defended, defended+DP, and attack model variants.

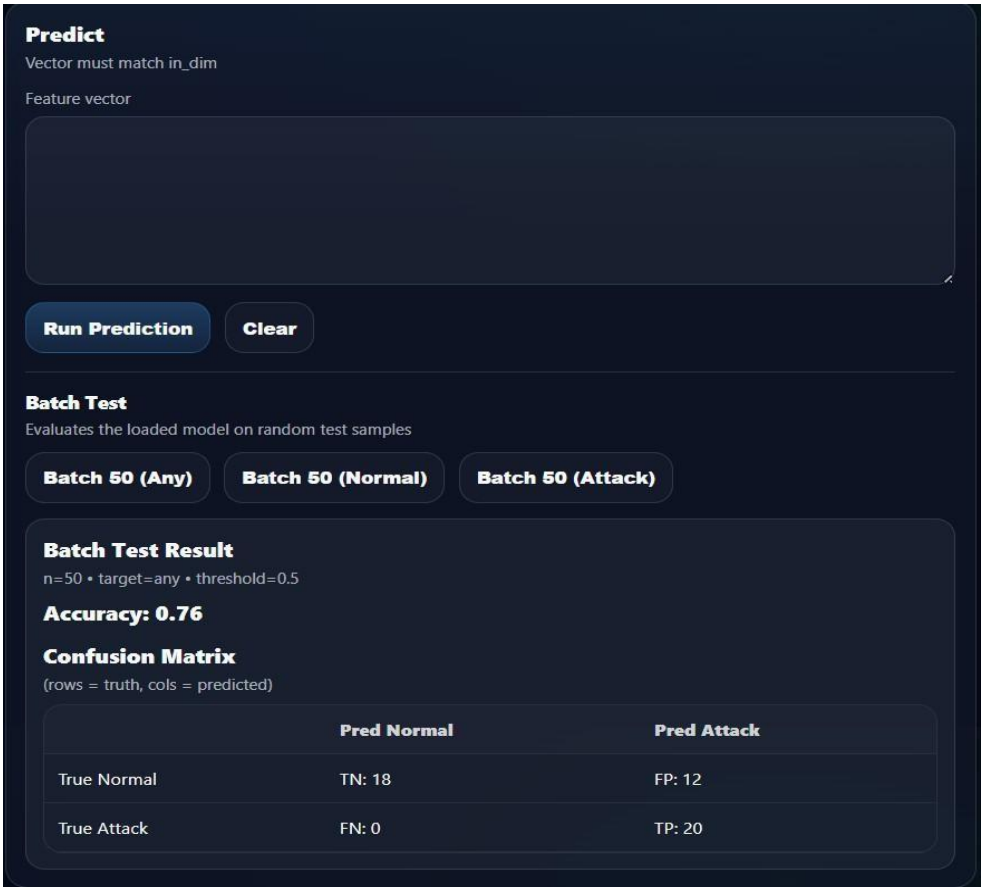


Figure 9.3. Batch-test result from the QSHIELD-FL inference dashboard showing accuracy and confusion-matrix output for the loaded SVD-hybrid rank-16 checkpoint.

9.1 Classical Reference Comparison

The centralized classical reference models provide an additional perspective on the final results. Their role is different from the federated scenario-based evaluation, because they do not include client-side update exchange, attack simulation, robust aggregation, or DP-style perturbation. However, they are still useful as conventional baselines for the same NSL-KDD binary classification task.

Among the three classical baselines, the Decision Tree achieved the strongest overall reference performance with an accuracy of 0.7833 and an F1-score of 0.7854, followed closely by the compact MLP with an accuracy of 0.7794 and an F1-score of 0.7834. Logistic Regression was weaker overall, with an accuracy of 0.7511 and an F1-score of 0.7621.

These results show that the proposed hybrid model is not competing against an unrealistically weak classical floor. At the same time, the comparison should be interpreted carefully: the classical models are centralized reference runs, whereas the main QSHIELD-FL system is evaluated in the more demanding federated setting with adversarial and privacy-aware scenarios. Therefore, the fairest conclusion is not that the hybrid model universally outperforms classical learning, but that it remains competitive with standard classical references while additionally supporting federated training, malicious-update simulation, robust aggregation, and privacy-oriented extensions.

Model Path	Accuracy	F1-Score	Comparison Role
Logistic Regression	0.7511	0.7621	Centralized classical reference
MLP	0.7794	0.7834	Centralized classical reference
Decision Tree	0.7833	0.7854	Centralized classical reference
Final SVD-Hybrid r16 (clean)	0.8072	_	Federated hybrid final model

10. Discussion

A few things came out clearly from the experiments. First, running a hybrid VQC inside a federated IDS pipeline is practically achievable. The model trains stably across rounds and does not collapse under the federated setup. Second, and not surprisingly, naive FedAvg is genuinely fragile. When a single malicious client runs SignFlip at scale 5.0, accuracy drops from around 80% to near chance. Third, Trimmed Mean largely recovers from the damage. Defended accuracy comes back to the high 70s across SVD variants, which validates using it as the primary defense.

The rank sweep turned out to be more informative than expected. Rank 32 actually did best under the attack only scenario (0.5661 vs 0.4308 for the others), which initially seemed like a reason to prefer it. But when defense was applied, rank 32 dropped to 0.7517, below both rank 16 and rank 24. What seems to be going on is that keeping more dimensions changes how the model absorbs poisoned updates under naive aggregation, but the advantage disappears once Trimmed Mean is doing its job. Since the whole point of the system is to work with defenses, rank 16 balanced behavior (0.8072 clean, 0.7741 defended plus DP) made it the sensible final choice.

A few honest limitations are worth stating plainly. The federated setup is a simulation: three or nine clients running on one machine, not actual distributed nodes. Differential privacy in the current code is Gaussian update noise with $\sigma = 0.003$; there is no gradient clipping or epsilon-delta accounting, so it is a practical approximation rather than a formal DP guarantee. QKD was part of the original project scope but is not implemented anywhere in the actual codebase. It is a future direction, not a current feature.

The centralized classical reference baselines also help interpret the final model more fairly. They show that the proposed hybrid path is not being compared only against trivial alternatives. The Decision Tree and compact MLP remain reasonably competitive on the binary NSL-KDD task, which is expected in a centralized tabular setting. At the same time, the final SVD-hybrid rank-16 model remains competitive with these references while also operating inside the broader federated, adversarial, and privacy-aware framework that defines the project's actual contribution.

None of the above breaks the project, but it matters for interpreting the numbers correctly. The system does what it claims to do within the bounds of a simulation based course project, and the four scenario evaluation gives a clearer picture of real behavior than a single clean accuracy number would.

11. Conclusion

This report presented the finalized version of QSHIELD-FL, a quantum federated intrusion-detection system with built-in adversarial defenses. The system integrates a VQC-based hybrid IDS model, a federated client-server training loop, SignFlip attack simulation, Trimmed Mean robust aggregation, DP-style update noise, and an SVD-enhanced hybrid learning extension with rank selection.

The final implementation and result analysis identify rank 16 as the best SVD setting for deployment because it provides the most balanced behavior across the clean, defended, and defended-plus-DP scenarios. The frontend and backend were then updated so that this finalized rank-16 path became the default demo configuration while baseline compatibility was preserved.

In addition, lightweight centralized classical reference baselines were evaluated to place the final hybrid results in context, showing that the proposed SVD-hybrid rank-16 path remains competitive with standard non-quantum models while addressing the broader federated, adversarial, and privacy-aware objectives of the project.

Compared with the earlier report, the current version improves the technical depth, the objective-to-method logic, the literature-to-design connection, the precision of the circuit explanation, the algorithmic detail, and the experiment-based model-selection story. Future work can extend the system through runtime QKD support, stronger formal differential privacy, repeated-seed studies, and true networked multi-node deployment.

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